The Columbia Glacier

An enormous Alaskan glacier that moves quickly but is also rapidly shortening in length

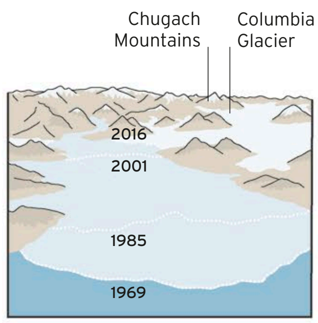
The Columbia Glacier is one of Alaska's largest glaciers, with an area of about 360 square miles (920 square km) and a length of 25 miles (40 km). It descends from the Chugach Mountains in southern Alaska to Prince William Sound (an inlet of the Gulf of Alaska), where it calves large numbers of icebergs into the sea.

Shrinking record

The Columbia Glacier is one of the fastest moving in North America, but its fast forward movement is offset by extremely rapid calving of icebergs at its terminus, at the rate of some 14 million tons (13 million tonnes) every day. As a result, since the 1960s, the glacier's terminus has moved backward, and since 2011 it has become detached from some of its largest tributary glaciers. The retreat cannot be attributed solely to global warming, as other nearby glaciers are not shrinking at the same pace. Instead, experts believe it is connected to the shape of the bedrock channel under the glacier.

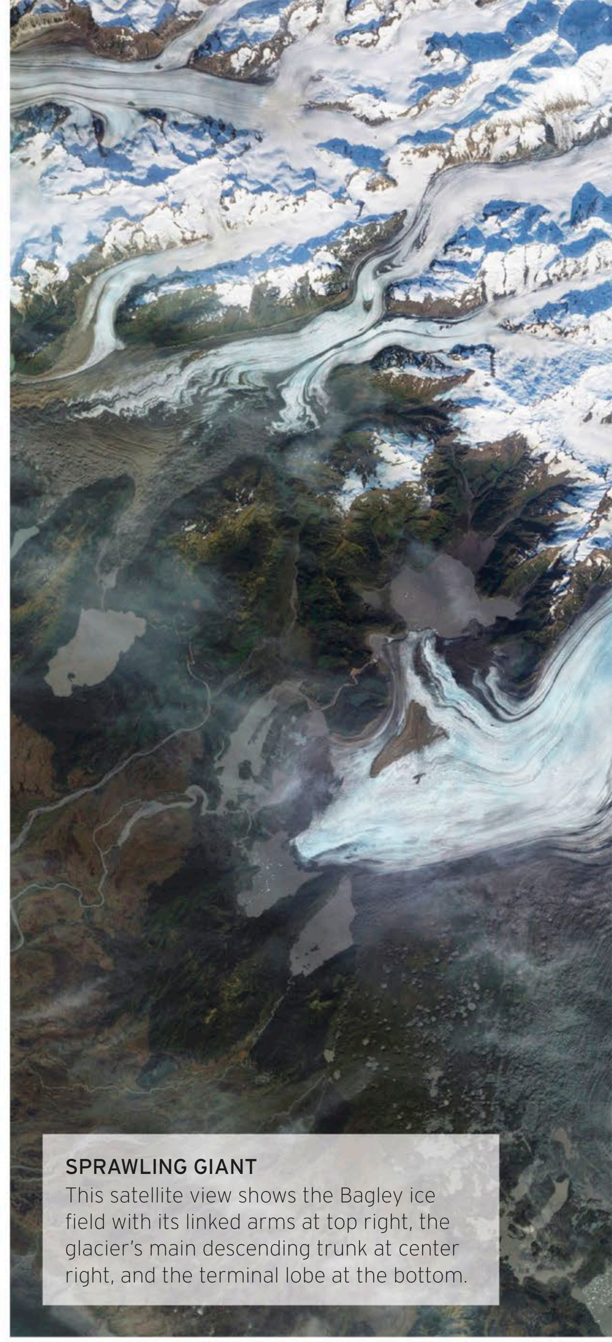
THE RETREAT OF THE COLUMBIA GLACIER

Since the 1960s, the iceberg-calving front of the Columbia Glacier has retreated about 12 miles (20 km). Marked here are its positions at roughly 15-year intervals since 1969.

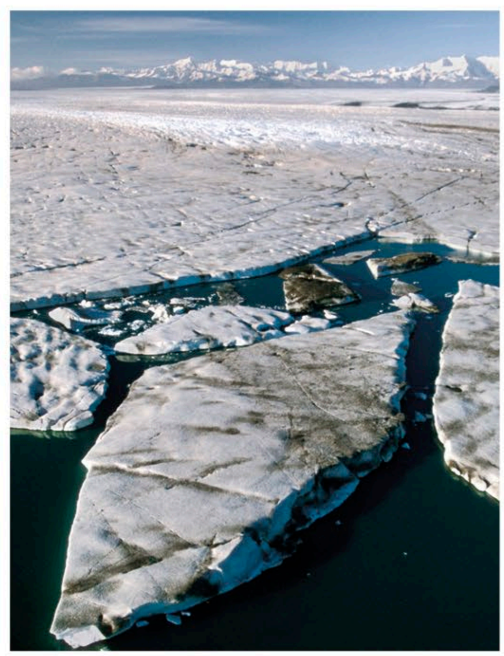


◁ **RETREATING TERMINUS**
About two-thirds of the area of the glacier visible in this photograph, which was taken in the mid-2000s, is now open water.

▽ **MILITARY FORMATION**
The higher part of the glacier is criss-crossed by crevasses, producing an army of ice pinnacles on its surface.



SPRAWLING GIANT
This satellite view shows the Bagley ice field with its linked arms at top right, the glacier's main descending trunk at center right, and the terminal lobe at the bottom.



△ **ICE SHARDS**
Massive icebergs regularly break off the end of the glacier and may then float around for months in Vitus Lake.